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PROJECT NAME

Insider Threat Radar – Behavior Analytics for Employee Risk Prediction

Client Project ID	MV_26_01
Team	Champs
Project Lead	ajay
Assigned To	Heena
Project Dead Line	08/01/2026

Project Requirements with Description and Outcome

Requirement	What to do	Outcome / Result
Insider threat dataset	Load CMU-CERT Insider Threat Dataset containing multi-log employee activity	Dataset becomes available for analysis

Data Analysis & EDA

Requirement	What to do	Outcome / Result
Head, tail, info, dtypes, shape	Inspect each log file	Understand data structure
User-wise activity distribution	Count events per employee	Identify heavy/light users
Time-based analysis	Analyze hourly/daily/weekly activity	Understand normal behavior patterns
Event frequency analysis	Count logins, downloads, emails per user	Detect unusual spikes
Missing values & duplicates	Identify nulls and repeated records	Data quality assessment
Cross-log comparison	Compare activity across multiple logs	Detect correlated behaviors

Data Cleaning

Requirement	What to do	Outcome / Result
Remove duplicates	Drop repeated log entries	Clean activity logs
Handle missing values	Impute or remove nulls	Consistent dataset
Timestamp normalization	Convert all timestamps to uniform format	Time-aligned logs
User log ordering	Arrange logs chronologically per employee	Sequential behavior tracking

Feature Engineering

Requirement	What to do	Outcome / Result
Login features	Extract login frequency, session duration, off-hour access	Behavioral login metrics



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File activity features	Compute file access count, download volume	Data access behavior
Email features	Extract number of recipients, CCs, body length	Communication behavior metrics
Content features	Apply TF-IDF on email content	Text-based behavior indicators
Web activity features	Count browsing frequency	Internet usage patterns
USB features	Count insert/remove events	External data transfer signals
Time-based features	Hour-of-day, day-of-week patterns	Temporal behavior representation

Anomaly Detection (Unsupervised)

Requirement	What to do	Outcome / Result
Train Isolation Forest	Learn normal employee behavior	Detect sudden rare anomalies
Generate anomaly scores	Score new activities	Quantify abnormal behavior
Threshold-based detection	Flag high anomaly scores	Identify suspicious events

Time-Series Behavior Modeling

Requirement	What to do	Outcome / Result
Prepare sequential data	Create daily/weekly sequences per user	Time-ordered behavior input
Train LSTM / GRU models	Learn long-term behavior patterns	Detect gradual insider buildup
Time-series anomaly scores	Generate deviation scores	Capture slow insider threats

Supervised & Hybrid Modeling

Requirement	What to do	Outcome / Result
Pseudo-label generation	Use anomaly outputs as weak labels	Enable supervised learning
Train Random Forest	Classify normal vs malicious behavior	Tree-based insider classifier
Train Gradient Boosting	Improve detection of complex patterns	Boosted insider classifier
Train MLP	Learn non-linear feature interactions	Neural network classifier
Stacking ensemble	Combine multiple models	Improved overall prediction accuracy



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Risk Scoring Module

Requirement	What to do	Outcome / Result
Assign severity weights	Define high/medium/low risk actions	Weighted anomaly contribution
Compute final risk score	Aggregate scores (0–100) per employee	Employee-wise threat ranking
Risk categorization	Low / Medium / High risk levels	Easy prioritization for admins

Visualization & Dashboard

Requirement	What to do	Outcome / Result
Anomaly timeline	Plot daily/weekly anomalies	Behavior trend visualization
Risk score charts	Display risk trends per employee	Quick risk assessment
User activity dashboard	Show logs, anomalies, scores	Centralized monitoring interface
Alert generation	Trigger alerts for high-risk users	Proactive security response

Evaluation

Requirement	What to do	Outcome / Result
Confusion matrix	Compare predicted vs actual labels	Error pattern analysis
Precision, Recall, F1-score	Evaluate minority-class detection	Insider threat detection quality
False positive analysis	Measure alert accuracy	Reduced false alarms
Model comparison	Compare RF, GB, MLP, LSTM, GRU, ensemble	Select best-performing model

UI

Requirement	What to do	Outcome / Result
Live log ingestion	Accept continuous employee activity logs	Real-time monitoring
Run anomaly + time-series + risk scoring	Execute full pipeline	Early insider threat prediction
Admin alert output	Display alerts with explanations	Actionable security intelligence